KINDERGARTEN · INTRO BLOCK CODING

Build Bit's Stack

Draw the blocks to get Bit to the star

Name: .... Date: ....

★ I can build a short program to get Bit to the star.

You build it

Now YOU make the program. Draw an arrow in each empty block so Bit reaches the star ★. Use ➡ to go right and ⬅ to go left. Then run your stack from the top to check Bit lands on the star.

Build 1 — Bit on box 1, star on box 4

1	2	3	4	5
Bit			★	

Your program

⚑

[ ]

[ ]

[ ]

Three empty blocks to fill.

1 Get to the star

Draw an arrow in each of the 3 empty blocks so Bit goes from box 1 to the star on box 4. Then run it to check.

Build 2 — Bit on box 4, star on box 2

1	2	3	4	5
	★		Bit	

Your program

⚑

The star is to the LEFT this time.

2 Go left to the star

Draw an arrow in each of the 2 empty blocks so Bit goes from box 4 to the star on box 2. Then run it to check.

Build 3 — Bit on box 1, star on box 3

1	2	3	4	5
Bit		★		

Your program

⚑

3 You try — build it yourself

Draw an arrow in each of the 2 empty blocks so Bit goes from box 1 to the star on box 3. Then run it to check.

Big idea

You wrote a program! You picked the blocks and put them in order to make Bit reach the goal. That is what coders do — then they run it to make sure it works.

TEACHER GUIDE · KINDERGARTEN · INTRO BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Children build their own stack — choosing and ordering move blocks — to make Bit reach the star, then run it to check. Turning a goal ("get to box 4") into an ordered set of blocks is the writing side of coding. No coding experience needed.

Before you start (no computer needed)

No computer — paper, a pencil, and a crayon. You read each prompt aloud. Optional: paper blocks (a green flag block + arrow blocks) to stack by hand, or floor boxes the child can walk.

How to lead it (about 15 minutes)

1. Show them (you do). Count the boxes from Bit to the star, then draw one arrow per box.
2. Do one together. Exercise 1 — box 1 to box 4 needs three right arrows.
3. Let them try. Children build a left-arrow stack (Ex 2) and a two-step stack (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Build 1: right, right, right (box 1 → 2 → 3 → 4, the star). Exercise 2 — Build 2: left, left (box 4 → 3 → 2, the star).

Exercise 3 — Build 3: right, right (box 1 → 2 → 3, the star).

If children get stuck — and ways to adjust

Watch for: too many or too few arrows. Count the boxes between Bit and the star first.

Make it easier: place a finger on each box and draw one arrow per box.

Make it harder: ask for a stack that starts somewhere else and still reaches the star.

Make it active: a child walks their built stack on the floor boxes to test it.

Ontario Kindergarten link (official wording)

Problem Solving and Innovating — children plan and create a sequence of steps to reach a goal, then test and refine it.

In plain terms: writing your own stack of blocks to reach the star.

No reading is needed from the child — you read each prompt aloud; the child circles, numbers, or draws an arrow.

You'll know it worked when

The child builds a stack with the right arrows and number of steps to reach the star.